

* Difference Finite Automata & Push down automata

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|--|---|
| ① FA is simple idealize machine used to recognize patterns within input taken from some character set. | ① PDA is a type of automata that employs a stack |
| ② FA does not have capability to store long sequence of alphabet. | ② PDA has stack to store the input alphabet |
| ③ FA can be constructed for type (3) grammar. | ③ PDA can be constructed for type (2) grammar |
| ④ Acceptance in FA of any string is at final state. | ④ Input alphabet are accepted by reaching
i) Empty stack
ii) final state. |
| ⑤ FA is constructed for regular language. | ⑤ PDA is constructed for context free language. |

ae) construct a PDA that accepts the language
 $L = \{w c w^R\}$ over $\Sigma = \{a, b, c\}$
 where $w = (a+b)^*$

4ⁿ $w c w^R$ where 'w' is string, 'c' is constant
 w^R - Reverse of the string

$L = \{a c a, b c b, a b c b a, b a c a b, a b b c b b a, \dots\}$

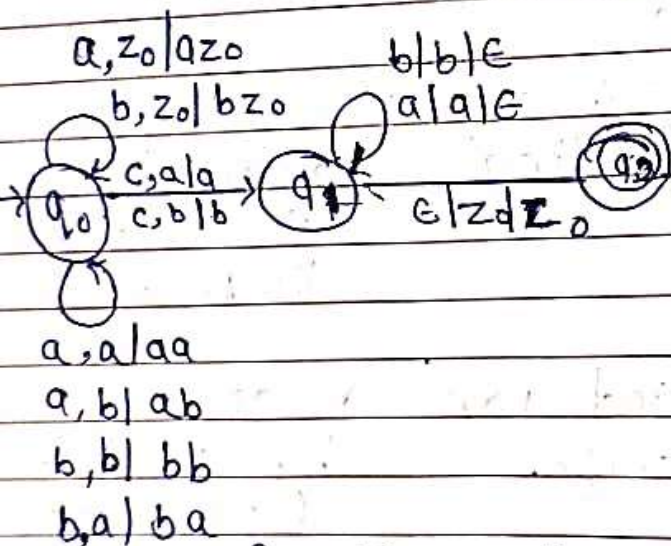
pushing a, b
 into stack
 upto constant
 c will not count

b
b
a
z ₀

b
b
a
z ₀

alter
 Round
 continue
 Reading

then b found
 pop 'b' from
 stack ok



Transition Function for string "abbcbbba"

$\delta(q_0, z_0, z_0) = (q_0, a z_0) \dots$ push a
 $\delta(q_0, a, b) = (q_0, a b) \dots$ push b
 $\delta(q_0, b, b) = (q_0, b b) \dots$ push b
 $\delta(q_0, c, b) = (q_1, b) \dots$ skip or Read
 $\delta(q_1, b, b) = (q_1, \epsilon) \dots$ pop 'b'
 $\delta(q_1, b, b) = (q_1, \epsilon) \dots$ pop b
 $\delta(q_1, a, a) = (q_1, \epsilon) \dots$ pop a
 $\delta(q_1, \epsilon, z_0) = (q_2, z_0) \dots$

Ques) construct a PDA that accepts the language
 $= \{a^n b^n c \mid n \geq 1\}$

Solution

- 1) No. of a's = no. of b's
- 2) a's followed by b
- 3) followed by c

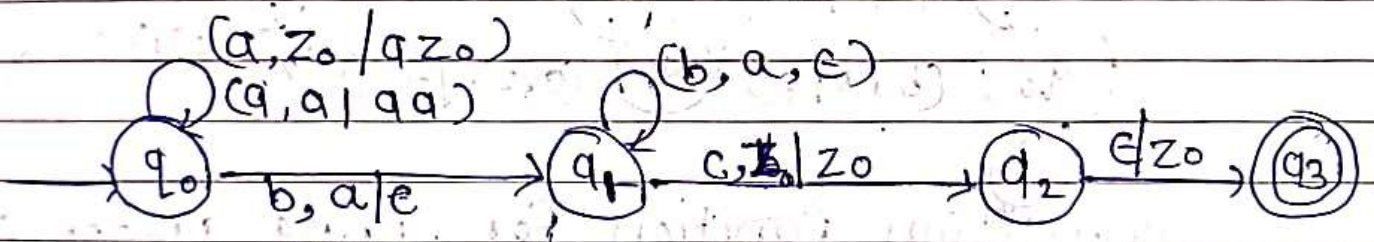
$L = \{abc, aabbc, aaabbbbc, \dots\}$

push 'a' into stack untill b is not found

a
a
a
z ₀

a
a
a
z ₀

push when found b



Transition function for string "aaabbbbc"

$\delta(q_0, a, z_0) = (q_0, a z_0)$... push a
 $\delta(q_0, a, a) = (q_0, a a)$... push a
 $\delta(q_0, a, a) = (q_0, a a)$... push a
 $\delta(q_0, b, a) = (q_1, \epsilon)$... pop a
 $\delta(q_1, b, a) = (q_1, \epsilon)$... pop a
 $\delta(q_1, b, a) = (q_1, \epsilon)$... pop a
 $\delta(q_1, c, z_0) = (q_2, z_0)$... Read the 'c'
 $\delta(q_2, \epsilon, z_0) = (q_3, z_0)$... final state

PDA is

$M = \{ \{q_0, q_1, q_2, q_3\}, \{a, b, c\}, \{a, z_0\}, z_0, \Phi, \delta \}$

Ques: construct a PDA that accepts the language
 $L = \{a^n b^n c^m \mid m, n \geq 1\}$

where $\Sigma = \{a, b, c\}$

Solution

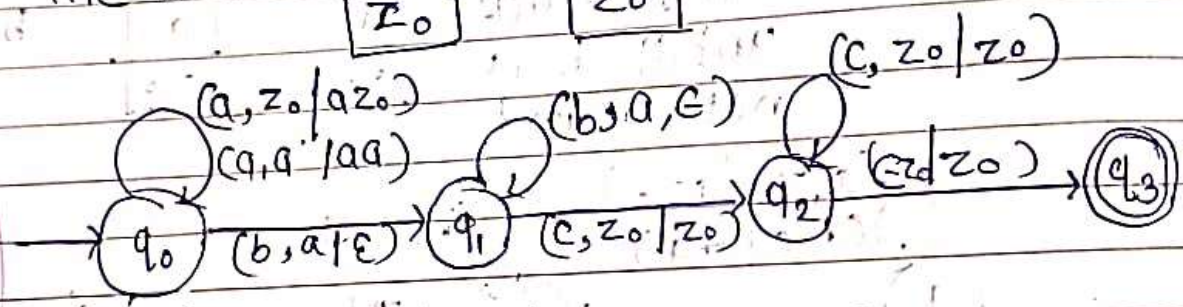
$L = \{abc, aabbcc, aaabbbccc, \dots\}$

push 'a' on top of the stack when 'b' found in string.

a
a
a
z_0

a
a
a
z_0

pop 'a'



Transition function for string aaabbbccc

- $\delta(q_0, a, z_0) = \delta(q_0, az_0)$
- $\delta(q_0, a, a) = \delta(q_0, aa)$
- $\delta(q_0, a, a) = \delta(q_0, aa)$
- $\delta(q_0, b, a) = \delta(q_1, \epsilon)$
- $\delta(q_1, b, a) = \delta(q_1, \epsilon)$
- $\delta(q_1, b, a) = \delta(q_1, \epsilon)$
- $\delta(q_1, c, z_0) = \delta(q_2, z_0)$
- $\delta(q_2, c, z_0) = \delta(q_2, z_0)$
- $\delta(q_2, c, z_0) = \delta(q_2, z_0)$
- $\delta(q_2, c, z_0) = \delta(q_2, z_0)$
- $\delta(q_2, \epsilon, z_0) = \delta(q_3, z_0)$

PDA is

$M = \{ \{q_0, q_1, q_2, q_3\}, \{a, b, c\}, z_0, \delta, \emptyset, \{z_0\} \}$

Equivalence of PDA's & CFG's.

Step 1: check grammar is in GNF or not. If not convert into GNF.
1) conversion of CFG to PDA:-

CFG consist production P. where variables is non-terminal symbols & terminal are terminals which available in CFG.

we have to follow two rules.

Rule 1: for each variable A

$\delta(q, \epsilon, A) = (q, \beta)$ where $A \rightarrow \beta$ is a production of grammar.

Rule 2:

for each terminal 'a'.

$\delta(q, a, a) = (q, \epsilon)$

eg: for the following grammar design PDA and test string "010000"

eg:

$S \rightarrow 0BB$
 $B \rightarrow 0S \mid 1S \mid 0$

Test: 010000

solution:

The equivalent PDA is

$\delta(q, \epsilon, S) = (q, 0BB) \dots ①$
 $\delta(q, \epsilon, B) = (q, 0S), (q, 1S), (q, 0) \dots ②$
 $\delta(q, 0, 0) = (q, \epsilon) \dots ③$
 $\delta(q, 1, 1) = (q, \epsilon) \dots ④$

$\delta(q, 010000, S)$ using transition 1
 $\delta(q, 010000, \emptyset BB)$... ③

↑ stack of top is 0
 so we use $\delta(q, 0, \emptyset)$... pop '0'
 ↑ stack top 0

so,

$\delta(q, 10000, BB)$... ②

$\delta(q, 10000, 1SB)$... ④

$\delta(q, 0000, SB)$... ①

$\delta(q, 0000, \emptyset BBB)$... ③

$\delta(q, 000, BBB)$... ②

$\delta(q, 000, \emptyset BB)$... ③

$\delta(q, 00, BB)$... ②

$\delta(q, 00, \emptyset B)$... ③

$\delta(q, 0, B)$... ②

$\delta(q, \emptyset, \emptyset)$... ③

$\delta(q, \epsilon, \epsilon)$ Accepted

The string accepted by PDA.

eg2

$S \rightarrow 0S1 \mid 00 \mid 11$

consider string '0111'

The equivalent PDA for the grammar

$\delta(q, \epsilon; S) \Rightarrow (q, 0S1)(q, 00)(q, 11)$

$\delta(q, 0, 0) = (q, \epsilon)$

$\delta(q, 1, 1) = (q, \epsilon)$... ③

- $\delta(q, 0111, S) \dots \dots \textcircled{1}$
- $\delta(q, 0111, 0SD) \dots \dots \textcircled{2}$
- $\delta(q, 111, SD) \dots \dots \textcircled{1}$
- $\delta(q, 111, 111) \dots \dots \textcircled{3}$
- $\delta(q, 11, 11) \dots \dots \textcircled{2}$
- $\delta(q, 1, 1) \dots \dots \textcircled{3}$
- $\delta(q, \epsilon, \epsilon) \dots \dots \text{accepted}$

The string '0111' is test by PDA is accepted same as CFG.

eg3 Draw a PDA for the CFG given below.

$$S \rightarrow aSb$$

$$S \rightarrow a | b | \epsilon$$

Solution:

The mapping function δ will be

- $\delta(q, \epsilon, S) = \delta(q, aSb) \dots \dots \textcircled{R_1}$
- $\delta(q, \epsilon, S) = \delta(q, a), (q, b), (q, \epsilon) \dots \textcircled{R_2}$
- $\delta(q, a, a) = (q, \epsilon) \dots \textcircled{R_3}$
- $\delta(q, b, b) = (q, \epsilon) \dots \textcircled{R_4}$

consider the string aaabb

- $\delta(q, aaabb, S) \dots \dots R_1$
- $\delta(q, aaabb, aSb) \dots \dots R_3$
- $\delta(q, aabb, Sb) \dots \dots R_1$
- $\delta(q, aabb, aSb b) \dots \dots R_3$
- $\delta(q, abb, Sbb) \dots \dots R_2$
- $\delta(q, abb, abb) \dots \dots R_3$
- $\delta(q, b, bb) \dots \dots R_4$

eg4: construct a PDA equivalent to following grammar:

$$S \rightarrow aAA$$

$$A \rightarrow as|bS|a$$

Solution: The mapping function δ of grammar

$$\delta(q, \epsilon, S) = \delta(q, aAA) \dots R_1$$

$$\delta(q, \epsilon, A) = \delta(q, as), (q, bS), (q, a) \dots R_2$$

$$\delta(q, a, a) = \delta(q, \epsilon) \dots R_3$$

$$\delta(q, b, b) = \delta(q, \epsilon) \dots R_4$$

consider string substitute: "aaaaaa"

$S \rightarrow aAA$	$\delta(q, aaaaaa, S) \dots R_1$
$A \rightarrow as bS a$	$\delta(q, aaaaaa, aAA) \dots R_2$
$S \rightarrow aAA$	$\delta(q, aaaaaa, AA) \dots R_2$
$S \rightarrow aaaaaa$	$\delta(q, aaaaaa, aSA) \dots R_3$
$S \rightarrow aaaaaa$	$\delta(q, aaaaaa, SA) \dots R_1$
$S \rightarrow aAA$	$\delta(q, aaaaaa, AAA) \dots R_2$
$S \rightarrow aAA$	$\delta(q, aaaa, AAA) \dots R_4$
$\rightarrow aaaS$	$\delta(q, aaaa, AA) \dots R_3$
$\rightarrow aaaaaAA$	$\delta(q, aa, AA) \dots R_4$
$\rightarrow aaaaaaA$	$\delta(q, a, AA) \dots R_3$
$\rightarrow aaaaaaa$	$\delta(q, a, A) \dots R_2$
	$\delta(q, a, a) \dots R_3$
	$\delta(q, \epsilon, \epsilon) \dots \text{accepted}$

2) conversion of PDA to CFG:

eg: $M = \{P, q\}, \{0, 1\}, \{\epsilon, x\}, \delta, q, z$ initial state symbol

- ① $\delta(q, 1, z) = (q, zz)$... Push
- ② $\delta(q, 1, x) = (q, xx)$... Push
- ③ $\delta(q, \epsilon, x) = (q, \epsilon)$... Pop
- ④ $\delta(q, 0, x) = (P, x)$... Nop
- ⑤ $\delta(P, 1, x) = (P, \epsilon)$... Pop
- ⑥ $\delta(P, 0, z) = (q, z)$... Nop.

Solution:

$S \rightarrow [q, z, q]$... initial state given
 $S \rightarrow [q, z, P]$... Possibility

① $\delta(q, 1, z) \Rightarrow (q, xz)$... push

$[q, z, q] \rightarrow 1[q, x, q] [q, z, q]$

$[q, z, q] \rightarrow 1[q, x, P] [P, z, q]$

$[q, z, P] \rightarrow 1[q, x, q] [q, z, P]$

$[q, z, P] \rightarrow 1[q, x, P] [P, z, P]$

② $\delta(q, 1, x) = (q, xx)$... push

$[q, x, q] \rightarrow 1[q, x, q] [q, x, q]$

$[q, x, q] \rightarrow 1[q, x, P] [P, x, q]$

$[q, x, P] \rightarrow 1[q, x, q] [q, x, P]$

$$\textcircled{3} \delta(q, \epsilon, x) \Rightarrow (q, \epsilon) \dots \text{POP}$$

$$[q, x, q] \rightarrow \epsilon$$

$$\textcircled{4} \delta(q, 0, x) \Rightarrow (p, x) \dots \text{NOP}$$

$$[q, x, q] \rightarrow 0 [p, x, q]$$

$$[q, x, p] \rightarrow 0 [p, x, p]$$

$$\textcircled{5} \delta(p, 1, x) \Rightarrow (p, \epsilon) \dots \text{POP}$$

$$[p, x, p] \rightarrow 1 \epsilon$$

$$\textcircled{6} \delta(p, 0, z) \Rightarrow (q, z) \dots \text{NOP}$$

$$[p, z, q] \rightarrow 0 [q, z, q]$$

$$[p, z, p] \rightarrow 0 [q, z, p]$$

Rename the production, we get final grammar

$$[q, z, q] = A$$

$$[q, z, p] = B$$

$$[p, z, q] = C$$

$$[p, z, p] = D$$

Final grammar

$$S \rightarrow A$$

$$S \rightarrow B$$

$$A \rightarrow 1EA$$

$$A \rightarrow 1FC$$

$$B \rightarrow 1EB$$

$$[q, x, q] = E$$

$$[q, x, p] = F$$

$$[p, x, q] = G$$

$$[p, x, p] = H$$

$$E \rightarrow 1EE$$

$$E \rightarrow 1FG$$

$$F \rightarrow 1EF$$

$$F \rightarrow 1FH$$

$$E \rightarrow \epsilon$$

$$H \rightarrow 1$$

$$C \rightarrow 0A$$

$$D \rightarrow 0B$$

summarize the grammar:

$S \rightarrow A | B$

$A \rightarrow 1EA | 1FC$

$B \rightarrow 1EB | 1FD$

$E \rightarrow 1EE | 1EG | \epsilon | OG$

$F \rightarrow 1EFF | 1FH | OM$

$H \rightarrow 1$

$C \rightarrow OA$

$D \rightarrow OB$

Assignment No. 4

① design pushdown automata (PDA)
for CFL
 $a^n b^{2n+1}$

② differentiate between finite automata
and pushdown automata.

③ Design pushdown automata for
 $a^n b^n$ where $n \geq 1$

④ convert following CFG to PDA.

$$S \rightarrow aSa$$

$$S \rightarrow bSb$$

$$S \rightarrow c$$

⑤ convert following CFG to PDA

$$S \rightarrow aS$$

$$S \rightarrow aA$$

$$A \rightarrow bA$$

$$A \rightarrow b$$